5. Write the Python Program for Missionaries Cannibal Problem

Reg. No: 192511013

Aim

To write a Python program to solve the Missionaries & Cannibals problem using the Breadth-First Search (BFS) algorithm.

Algorithm

1. Initially, Place three missionaries and three Cannibals on the left bank.
2. A boat Can Carry one or two people at a time.
3. Never allow Cannibals to outnumber missionaries on either bank when missionaries are present.
4. Generate all possible boat movement.
5. Reject invalid or previously visited states.
6. Called and Continue until all missionaries and Cannibals reach the right bank safely.
7. Display the sequence of states.

Program

```
from collection import deque
def valid (m, c):
    mr, cr = 3-m, 3-c
    return (0 <= m <= 3 and 0 <= c <= 3 and
            (m == 0 or m >= c) and
            (mr == 0 or mr >= cr))
```

```
def solve():
    start = (3,3,0)
    goal = (0,0,1)

    moves = [(1,0),(2,0),(0,1),(0,2),(1,1)]

    queue = deque([(start,[start])])
    visited = {started}

    while queue:
        state,path = queue.popleft()
        if state == goal:
            print("Solution steps:")
            for m,c,boat in path:
                side = "Right" if boat else "left"
                print("Missionaries:",m,
                      "Cannibals:",c,
                      "Boat:",side)
            return
        m,c,boat = state
        for dm,dc in moves:
            if boat==0:
                nm,nc = m-dm,c-dc
                nb = 1
            else:
                nm,nc = m+dm,c+dc
                nb = 0
            new_state = (nm,nc,nb)
```

```
        if value (nm, nc) and new_state not in visited:
            visited.add(new_state)
            queue.append(new_state, path + [new_state])
    Print("No Solution found.")
solve()
```

Sample Input

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Sample Output

Solution steps:

Missionaries: 3 Cannibals: 3 Boat: left
Missionaries: 3 Cannibals: 1 Boat: Right
Missionaries: 3 Cannibals: 2 Boat: left
Missionaries: 3 Cannibals: 0 Boat: Right
Missionaries: 3 Cannibals: 1 Boat: left
Missionaries: 1 Cannibals: 1 Boat: Right
Missionaries: 2 Cannibals: 2 Boat: left
Missionaries: 0 Cannibals: 2 Boat: Right
Missionaries: 0 Cannibals: 3 Boat: left
Missionaries: 0 Cannibals: 1 Boat: Right
Missionaries: 0 Cannibals: 2 Boat: left
Missionaries: 0 Cannibals: 0 Boat: Right

Result

Thus, the Missionaries and Cannibals problem was solve successfully using BFS.